# LoopCraft

## Closed-Loop AI Support Optimization Engine Powered by Exasol Personal

**Event:** ExaThon India 2026 · The Exasol Open Build Challenge

Track 04: The Work Behind the Work | Track 02: Customers, Channels & Demand

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# The Problem: The Support Feedback Black Hole

- Support AI agents handle millions of customer conversations, but when users give a thumbs-down, the failure signal is silently discarded.
- Support leads and knowledge authors cannot diagnose why automated conversations fail because feedback telemetry sits trapped across unlinked ticket logs.
- Knowledge base articles decay over time. As products and policies change, outdated KB articles mislead AI agents and cause repeat customer escalations.
- Today's ticket systems are optimized for transactional CRUD, not high-velocity analytical reasoning over unstructured conversational feedback.

# Problem Statement (Official Exasol Template)

**CORE CHALLENGE**

Support operations leads and knowledge engineers cannot diagnose why automated support conversations fail and proactively fix outdated guidance because negative customer feedback and conversational failure signals sit discarded across millions of unlinked support logs.

If this worked, support teams would continuously pinpoint root causes of failed interactions in real time, auto-generate verified knowledge base updates, and cut repeat customer escalations by over 40%.

# The Solution: LoopCraft

- Continuous Closed Loop: Intercepts negative ratings, classifies the root cause, and auto-drafts verified corrections.
- Automated KB Health Auditing: Continuously measures deflection and escalation rates across all articles.
- Multimodal Voice Briefing: Spoken audio briefs powered by ElevenLabs so admins can review failures hands-free.
- Powered by Exasol Personal: In-memory columnar analytics deliver sub-millisecond querying over tens of thousands of conversations.

# Why Exasol Personal is the Key Enabler

| Traditional Transactional DBs | Exasol Personal In-Memory Engine |
| --- | --- |
| ✕ Row-oriented storage slow for analytical scans<br>✕ Complex aggregation across millions of rows causes query timeouts<br>✕ Support feedback data remains locked in unanalyzed silos<br>✕ Slow batch ETL jobs delay KB fixes by weeks | ✓ Columnar in-memory execution delivers answers in < 5 milliseconds<br>✓ Instantly clusters failure types (Stale Article, No Coverage, Wrong Intent)<br>✓ Dynamic deflection trend calculation across 50,000+ interactions<br>✓ Live synchronization from raw support logs to actionable knowledge fixes |

# Architecture: The 4-Agent Closed Loop

- 1. Rating Watcher Agent: Intercepts customer thumbs-down events and logs telemetry directly to Exasol.
- 2. Failure Classifier Agent: Analyzes user query vs bot response, tagging failure type (Stale Article / No Coverage / Wrong Intent).
- 3. KB Author Agent: Synthesizes resolution notes from resolved human tickets to auto-draft updated articles.
- 4. Correction Brief Agent: Packages root cause diagnosis, spoken brief, and one-click approval into the admin queue.

# Technical Benchmarks: In-Memory Columnar Speed

- Synthetic Dataset: 50,000 conversational interaction logs and support tickets generated conforming to open-source rules.
- Category Deflection Scan: Aggregated across 50,000 rows in 21.9 ms.
- Root-Cause Failure Clustering: Filtered 17,484 negative interactions in 6.6 ms.
- Sub-Millisecond Dashboard Queries: Real-time UI widgets query Exasol views in < 4.5 ms.
- Zero Lag for IT Admins: Interactive triage dashboard with instantaneous responsiveness.

# Real-World Business Impact & ROI

**+44.3%**
Deflection Rate Jump on Outdated Articles (23.1% -> 67.4%)

**< 5 ms**
Exasol In-Memory Analytics Latency across 50k records

**100%**
Closed-Loop Feedback: Zero discarded negative ratings

**1-Click**
From Diagnosed Failure to Published KB Update

# ExaThon India 2026 Alignment

- Track 04: The Work Behind the Work — Transforming internal IT support and automated knowledge systems.
- Track 02: Customers, Channels & Demand — Mining churn and dissatisfaction signals sitting unused in support tickets.
- Exasol Personal Utilization: Core data warehouse and analytical engine for telemetry storage, aggregations, and views.
- Open Source Standards: MIT License, public GitHub repo, synthetic data only, Docker Compose quickstart.

# Deliverables & Links

- Public Code Repository: github.com/kavindon2007/fresh_works_kavin
- Live Web Demonstration: fresh-works-kavin.vercel.app
- Docker Compose Integration: Official exasol/docker-db setup with init schemas
- Open-Source License: MIT License included
- Pitch Deck & Demo Video Script: Included in repository docs